

ANNAPOLIS US NAVAL ACADEMY, MD, USA

WMO: 722158

Lat: 38.991N

Lon: 76.489W

Elev: 7

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	21.1	0.0	5.5	20.0	4.0	6.8	23.7	23.4	31.2	20.5	32.5	8.7	290	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	91.3	76.4	89.1	75.7	86.8	74.9	79.6	86.8	78.2	85.3	77.1	84.0	7.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	143.0	83.6	76.0	135.9	82.4	74.9	130.7	81.5	43.0	86.5	41.6	85.0	40.4	84.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	17.0	15.5		DB	12.4	96.5	4.8	3.2	8.9	98.8	6.1	100.7	3.4	102.5	-0.2	104.8
				WB	10.1	82.2	4.5	1.8	6.9	83.5	4.2	84.5	1.7	85.5	-1.6	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	36.5	38.1	45.3	56.4	65.8	74.9	79.4	77.6	72.2	60.5	49.3	41.4	(6)
	DBStd	16.76	8.91	8.91	8.64	7.57	6.99	5.21	4.30	4.17	5.88	7.53	7.30	7.82	(7)
	HDD50	1385	426	346	197	23	0	0	0	0	0	7	99	287	(8)
	HDD65	3996	883	753	610	274	77	3	0	0	8	182	473	733	(9)
	CDD50	4387	8	13	53	217	491	746	912	855	665	332	77	18	(10)
	CDD65	1522	0	0	0	17	103	299	447	390	224	41	1	0	(11)
	CDH74	11981	0	3	9	119	518	2182	4351	3270	1377	149	3	0	(12)
	CDH80	3513	0	0	0	31	111	596	1515	933	305	22	0	0	(13)
(14) Wind	WSAvg	6.7	7.0	7.4	7.9	7.8	6.7	6.3	5.8	5.7	6.1	6.4	6.5	6.3	(14)
(15) Precipitation	PrecAvg	46.0	3.5	2.6	4.3	3.6	3.7	4.2	4.4	4.5	4.3	3.9	3.1	3.7	(15)
	PrecMax	65.9	6.9	7.7	9.3	6.6	8.2	12.7	8.5	8.1	12.3	8.2	6.5	8.2	(16)
	PrecMin	35.3	1.4	0.3	0.0	1.5	0.7	2.2	1.6	1.2	0.6	0.1	0.6	1.4	(17)
	PrecStd	6.9	1.3	1.7	1.9	1.2	2.0	2.4	2.0	1.9	2.7	2.4	1.9	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	68.6	74.6	84.1	88.4	92.8	96.2	93.0	90.5	82.7	71.2	66.1	(19)
		MCWB	57.6	57.8	59.9	64.3	74.2	75.6	78.5	76.6	76.7	72.6	63.2	60.1	(20)
	2%	DB	56.3	60.3	66.8	76.8	82.7	89.0	92.2	89.8	86.1	78.2	66.0	60.0	(21)
		MCWB	50.2	51.7	56.1	62.5	70.4	74.6	77.4	76.1	74.9	70.5	58.8	55.8	(22)
	5%	DB	52.3	54.3	62.0	72.2	79.6	86.0	89.8	87.5	83.1	74.8	63.0	55.9	(23)
		MCWB	47.1	47.8	53.4	61.0	69.1	73.4	76.5	75.4	73.5	68.3	56.4	51.8	(24)
	10%	DB	48.3	49.8	57.2	68.2	76.3	83.3	87.4	84.6	81.0	72.1	60.2	52.2	(25)
		MCWB	43.5	44.0	49.5	58.9	67.6	72.4	75.7	74.2	71.8	65.5	54.5	47.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.5	62.1	69.3	75.9	79.0	82.3	81.1	78.9	75.1	64.7	62.0	(27)
		MCDB	65.8	66.4	70.7	74.3	83.9	87.3	90.5	87.0	86.7	79.4	68.5	65.1	(28)
	2%	WB	52.5	53.6	58.3	65.2	73.1	77.3	80.2	78.9	76.9	72.6	61.7	56.9	(29)
		MCDB	55.5	58.3	65.0	73.9	81.0	84.7	87.5	85.3	83.4	77.1	64.8	59.4	(30)
	5%	WB	47.9	48.5	54.9	62.6	71.1	76.0	78.7	77.6	75.6	69.9	58.4	52.6	(31)
		MCDB	51.3	52.9	60.8	70.4	77.5	83.1	85.9	84.2	81.7	73.9	61.9	54.8	(32)
	10%	WB	44.0	44.4	51.0	60.1	69.2	74.8	77.5	76.3	73.9	66.7	55.6	49.2	(33)
		MCDB	47.9	49.4	55.6	66.9	74.9	81.4	84.7	82.7	79.3	70.8	58.9	51.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.2	14.0	15.4	13.9	13.2	12.6	12.0	11.8	13.0	12.9	11.8	(35)
		MCDBR	18.7	20.5	21.3	21.9	16.7	15.8	15.1	14.4	13.5	14.3	16.5	16.7	(36)
	5% WB	MCWBR	14.8	14.9	13.9	11.3	8.8	7.9	7.0	6.8	6.9	8.9	11.8	14.6	(37)
		MCDWR	17.6	19.2	19.7	19.5	15.0	13.9	13.2	12.6	12.2	12.3	14.4	15.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.360	0.399	0.443	0.496	0.518	0.512	0.424	0.362	0.326	0.311	(40)
		taud	2.481	2.441	2.389	2.300	2.210	2.107	2.059	2.062	2.282	2.457	2.516	2.522	(41)
	Ebn at Noon		274	287	285	279	267	252	245	243	260	268	267	266	(42)
		Edh at Noon	25	30	35	40	45	50	52	51	38	29	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		661	930	1236	1577	1775	1966	1936	1719	1397	1041	753	580	(44)
	RadStd		52	104	97	122	212	150	93	97	156	113	64	55	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+142

Nomenclature: See separate page